EXPERIMENT 3

AIM: Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.

SOLUTION:

Aggregate functions are also called **group functions**.
They are **mathematical functions** that work on a set of rows and return **one single result**.

The main aggregate functions are:

- **COUNT** – used to count the number of rows
- **MAX** – used to find the highest value in a column
- **MIN** – used to find the lowest value in a column
- **SUM** – used to find the total of values in a column
- **AVG** – used to find the average of values in a column

Group By Clause:

The Group by statement/clause groups rows that have same values. i. e it is used to grouping the same type of values.

Or to group the data on the basis of some specific attributes/columns

- Group by clause used with aggregate functions like max, min, count, avg, sum etc.

Rule: Which column name is used after group by clause that column name should be written after select keyword.

Syntax:

SELECT column_name, AGGREGATE_FUNCTION(column_name)
FROM table_name
GROUP BY column_name;

Having Clause:

Having clause is used with group by only. In grouping we cann't use "where" clause so instead of "where" we will use Having Clause.
Or
To select the group on the basis of specified condition

Syntax:

SELECT column_name, AGGREGATE_FUNCTION(column_name)
FROM table_name
GROUP BY column_name
HAVING condition;

Order By clause:

Used to sort the result of SQL Query over the specified attribute.
SELECT column_name
FROM table_name
ORDER BY column_name [ASC | DESC];

The general syntax of SELECT command is:

Syntax:

SELECT [DISTINCT] column1, column2, ...

FROM table1, table2, ...
WHERE condition
GROUP BY column_name(s)
HAVING condition
ORDER BY column_name(s);

COUNT():
Query: Write a query to count the records in the student table.
SQL> SELECT COUNT(*) FROM student;
Output:?
Query: Write a query to count the number of distinct departments in the student table.
SQL> SELECT COUNT(DISTINCT deptno) FROM student;
Output:?
MAX():
Query: Write a query to find the maximum/highest marks in the student table
SQL> SELECT MAX(marks) FROM student;
Output:?
Query: Write a query to find the maximum/highest marks in the student table in IT department
SQL> SELECT MAX(marks) FROM student WHERE deptno = 10;
Output:?
MIN():
Query: Write a query to find the minimum/lowest marks in the student table
SQL> SELECT MIN(marks) FROM student;
Output:?
AVG():
Query: Write a query to find the average marks of students
SQL> SELECT AVG(marks) FROM student;
Output:?
SUM():
Query: Write a query to find the total/sum of marks of students
SQL> SELECT SUM(marks) FROM student;
Output:?
<u>GROUP BY:</u>
Query: Write a query to count the number of students in each department
SQL> SELECT deptno, COUNT(rollno)
FROM student
GROUP BY deptno;
Output:?
Query: Write a query to find the total (sum) of marks of students in each department
SQL> SELECT deptno, SUM(marks)
FROM student
GROUP BY deptno;
Output:?
Query: Write a query to find the average marks of students in each department
SQL> SELECT deptno, AVG(marks)
FROM student
GROUP BY deptno;
Output:?
Query: Write a query to display departments having more than 2 students

SQL> SELECT deptno, COUNT(rollno)
FROM student
GROUP BY deptno
HAVING COUNT(rollno) > 2;

Output:?

Query: Write a query to display departments whose total marks are greater than 150

SQL> SELECT deptno, SUM(marks
FROM student
GROUP BY deptno
HAVING SUM(marks) > 150;

Output:?

Query: Write a query to find sum, average, maximum, and minimum marks of each department

SQL> SELECT deptno,
SUM(marks) AS total_marks,
AVG(marks) AS avg_marks,
MAX(marks) AS max_marks,
MIN(marks) AS min_marks
FROM student
GROUP BY deptno;

Output:?

<u>Creation and dropping of Views:</u>

Views in SQL are considered as a virtual table.
A view also contains rows and columns like a real table in the database.
We can create a view by selecting fields from one or more tables present in the database.
A view can either have all the rows of a table or specific rows based on certain condition.

<u>Creating views:</u>

We can create a view using “create view” statement.
A view can be created from a single table or multiple tables.

Syntax:(Single table)

CREATE VIEW view_name AS
SELECT col1, col2
FROM table_name
WHERE condition;

Query: Write a query to create a view of IT department students

SQL> CREATE VIEW it_students As
SELECT rollno, name
FROM student
WHERE deptno = 10;

Output:?

Syntax: (Multiple tables)

Create view view_name as
Select s.name, d.dname , s.marks
From student s, dept d
Where s.deptno= d.deptno;

Query: Write a query to create a view showing student name, department name, and marks

SQL> CREATE VIEW student_dept_view AS
SELECT s.name, d.dname, s.marks
FROM student s, dept d

WHERE s.deptno = d.deptno;

Output:?

<u>Dropping a View:</u>

<u>Syntax:</u>

DROP VIEW view_name;

Query: Write a query to drop a view

SQL> DROP VIEW it_students;

Output:?